

AN2DL - Second Challenge Report

Bobert

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1 Introduction

This project addresses a supervised *histopathological image classification* problem using low-magnification Whole-Slide Images (WSI) samples from the **Grumpy Doctogres Dataset**. Each sample consists of a high-resolution histological image, and the objective is to assign the correct class label corresponding to the specific *molecular subtype* of the diseased tissue. The dataset comprises four classes: **Luminal A**, **Luminal B**, **HER2-enriched**, and **Triple Negative**. These classes represent distinct molecular profiles, which are known to correlate with different morphological and organizational patterns at the tissue level. The main objective is to design and evaluate a neural network architecture able to capture relevant visual patterns in the images. To obtain the best possible performance, we followed a structured strategy: (i) conducting data analysis to understand the dataset and derive implications for model design, (ii) pre-processing data to prepare the dataset for training, (iii) building the most suitable architecture for the task, and (iv) tuning the network parameters to improve generalization.

2 Problem Analysis [1]

The distribution of classes in the training set is unbalanced, with the *triple negative* category repre-

sented by fewer samples. This disparity can lead to a biased model behavior and therefore motivated us to use **weighted F1-score** as the evaluation metrics.

Further analysis revealed the presence of several **non-informative or corrupted samples** in the training set which, if not removed, would negatively affect the model training process. These include images affected by green stains or marker pen traces, as well as samples containing irrelevant content despite having apparently valid masks (e.g., images of the beloved Shrek). In addition, some samples contain more than one distinct tissue region within a single image, separated by a black background. Finally, the dataset exhibits significant variability in image spatial resolution and size, which is not ideal for training deep learning models and requires explicit preprocessing strategies.

3 Method

3.1 Data Preprocessing

Outliers removal An initial outliers removal strategy was based on supervised learning. Two separate binary Convolutional Neural Network (CNN) classifiers were trained: one to detect images corrupted by green stains [2] and one to identify samples containing irrelevant content (Shrek images) [3]. Both classifiers were based on a **ResNet-18** ar-

chitecture pretrained on **ImageNet**, chosen for its robustness and general-purpose visual feature extraction capabilities [4]. All images were loaded, padded and resized to a fixed spatial resolution compatible with the network input. Standard normalization was applied using the ImageNet mean and standard deviation, consistent with the pre-trained weights. Despite this setup, the classifiers exhibited poor generalization performance, indicating that the approach was unnecessarily complex.

A subsequent analysis showed that the outliers were characterized by **highly distinctive color patterns**. Green stains exhibited a saturated green tone never present in the original fabrics, while Shrek images showed a characteristic green-brown color gradient. Based on this observation, the outlier detection pipeline was reformulated as a color-based filtering problem [5] [6]. This approach ended up achieving **near-perfect outlier removal performance**.

Creating patches for each image [7] Given the issues highlighted in the problem analysis, we opted for a **patch-based analysis** strategy instead of processing whole images directly. By focusing on extracting patches guided by the provided binary masks, the model is encouraged to concentrate on tissue areas that are more likely to be informative. After obtaining predictions for all patches extracted from a given image, the corresponding patch-level outputs are supposed to be aggregated in order to infer the class label of the entire image. Patch extraction was performed using a sliding-window approach over the masked regions. For each window, a score was computed based on the proportion of foreground pixels within the binary mask, and only windows exceeding a predefined threshold were retained. For each image, a minimum of one and a maximum of three to five patches were extracted, while explicitly avoiding spatial overlap in order to prevent redundant information across patches. Different patch sizes (128×128 , 224×224 , and 256×256) were evaluated. The choice of patch resolution was driven by the backbone architecture employed, balancing the need to preserve relevant tissue context against the input size constraints of the selected models.

3.2 Data Augmentation

Due to the limited size of the dataset and the risk of overfitting, data augmentation was employed as a regularization strategy to promote invariance to non-discriminative geometric and photometric variations. Standard automated augmentation policies were evaluated, such as (*RandAugment*, *TrivialAugment*, *AutoAugment*), as well as a custom pipeline. The latter includes **geometric transformations** such as random horizontal flipping, rotations, translations, and isotropic scaling, together with **mild photometric perturbations** limited to brightness and saturation not to alter the problem’s domain.

More aggressive color-based transformations, including hue jittering, were intentionally excluded to avoid distorting staining-related information potentially relevant for molecular subtype discrimination. Similarly, *MixUp* was implemented and tested, but ultimately not adopted, as mixing patches from different images may introduce unrealistic patterns and amplify the label noise (as shown in Section 4). To prevent augmentation-induced artifacts at image boundaries, the empty regions created by geometric transformations were filled using the global mean color computed over the whole training set [1].

At inference time, **Test-Time Augmentation** (TTA) was applied to improve prediction robustness. Multiple augmented variants were generated for each patch using a restricted set of transformations consistent with training, and the corresponding predictions were aggregated by averaging softmax probabilities to obtain the final output. Finally, the patches are aggregated by image using either max confidence or a mean weighted on how much mask is present in the actual patch. The final aggregation is what in the end determines the prediction on the test.

4 Experiments

All models reported are the ones that achieved the best results using a specific architecture. The comparison is made using F1-score in validation. Test F1-score is not considered, the test results are subject to a certain degree of randomness due to the training process and the use of TTA.

Table 1: Best models comparison.

Model	Val-F1
EfficientNetB2 ¹	0.3033
ResNet50 ²	0.3779
CTrans ³	0.5060
CTrans + MixUp ⁴	0.4715

We initially evaluated convolutional architectures pretrained on *ImageNet* and commonly adopted in medical imaging, starting from **EfficientNet-B2** (≈ 8 M parameters) [8] as a lightweight and parameter-efficient baseline. We then explored deeper architectures such as **ResNet-50** (≈ 23 M parameters) [4] to assess whether increased model capacity and residual connections could better capture discriminative patterns.

Finally, we investigated hybrid CNN–Transformer architectures, specifically **CTrans** (≈ 27 M parameters) [9], which is pretrained on histopathological data and integrates attention mechanisms to model long-range spatial dependencies.

Regarding optimization, we experimented with the **Ranger** optimizer but observed less stable convergence and inferior validation performance compared to AdamW. We therefore retained **AdamW**, which provided more consistent behavior in the small-data fine-tuning regime.

5 Results

The best performance was achieved by the **CTrans** model without MixUp, reaching a test score of **0.4256**. An analysis of the experimental settings highlights several consistent trends: reduced fine-tuning of the backbone, lower learning rates, custom data augmentation, and input resolutions aligned with the pretrained architecture consistently led to improved performance.

6 Discussion

The performance achieved by the final model shows that we managed to reach a satisfying solution, obtaining a good generalization of the given problem. The extensive focus put on analyzing and cleaning the training data was essential, and payed off once the work on the actual classifier model began. Moreover, the implementation of TTA allowed us to reach competitive results on the test set. The main problem of the final model, as shown by the training process and plot [10], is a certain degree of overfitting to the training set. Finally, the solution proposed has some clear limitations, mainly due to a lack of extensive hyperparameters tuning and exploration of new preprocessing and augmentations techniques.

7 Conclusions

This work explored a histopathological image classification task through careful data analysis, patch-based preprocessing, and transfer learning. The use of domain-adapted pretrained models, custom data augmentation, and test-time augmentation proved effective, with CTrans achieving the best performance. However, residual overfitting and limited hyperparameter exploration remain open issues. Notably, a dataset with uniform image resolution and images already centered on informative tissue regions would likely yield improved performance by providing more resolved and discriminative visual information. Future work would focus on systematic hyperparameter optimization through grid or random search, further investigation of data augmentation strategies, and the exploration of alternative domain-specific pretrained models and aggregation schemes to improve generalization.

¹Patches 256x256. Two layers unfrozen. Learning Rate (classifier) = 1e-3, Learning Rate (fine tuning) = 1e-4, Dropout Rate = 0.3, Batch Size: 64, Ranger Optimizer, Custom Augmentation. [11]

²Patches 128x128. One layer unfrozen. Learning Rate (classifier) = 1e-3, Learning Rate (fine tuning) = 1e-4, Dropout Rate = 0.2, Weight Decay = 0.01, Label Smoothing = 0.05, Batch Size: 64, Ranger Optimizer, Custom Augmentation. [12]

³Patches 224x224. One layer unfrozen. Learning Rate (classifier) = 5e-5, Learning Rate (fine tuning) = 5e-4, Dropout Rate = 0.3, Weight Decay = 5e-4, Label Smoothing = 0.05, Batch Size: 64, AdamW Optimizer, Custom Augmentation. [10]

⁴Patches 224x224. One layer unfrozen. Learning Rate (classifier) = 5e-5, Learning Rate (fine tuning) = 5e-4, Dropout Rate = 0.3, Weight Decay = 5e-4, Label Smoothing = 0.05, Batch Size: 64, AdamW Optimizer, Mixup Alpha = 0.2, Custom Augmentation. [13]

References

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